



CareCaller Hackathon 2026

AI Call Quality & Voice Agent Challenge | March 19, 2026

992	88	8	2
Total Calls Analyzed	Review Tickets (Ground Truth)	Call Outcome Categories	Problem Statements

Background

CareCaller is an AI-powered healthcare validation call system. AI voice agents make outbound calls to patients for medication refill check-ins, collecting health questionnaire responses via natural conversation. Each call is transcribed, verified, and stored with structured outcomes. A small percentage of calls have quality issues that require human review -- these are tracked as **review tickets**. Your challenge is to build systems that automatically detect these issues and/or improve the voice agent experience.

What You Get

992 synthetic calls with full transcripts, structured Q&A responses (14 health questions), call metadata, and 88 review tickets as ground truth. All data is **fully synthetic** -- generated by AI with no connection to real patients. Health data (weights, medications, side effects) is fabricated but realistic.

Data Format

All splits are provided in three formats:

csv/ -- one file per split, flat tabular format

json/ -- one file per split, nested with metadata

parquet/ -- one file per split, columnar binary format

Use whichever format works best for your tools.



Problem 1: Call Quality Auto-Flagger

Build a system that automatically detects calls requiring human review.

Objective

Given a call's metadata, transcript, and Q&A responses, predict whether the call should be flagged for human review (**has_ticket = True**). This is a **binary classification** task with a ~9% positive rate -- the challenge is catching subtle anomalies across different issue types.

What Counts as a Ticket?

Outcome miscategorization -- Call marked as wrong outcome (e.g., patient opted out but call says 'completed')

STT mishearing -- Speech-to-text errors on critical values (e.g., weight '62' instead of '262')

Agent skipped questions -- AI agent marked call complete but missed required health questions

Wrong number misclassification -- Patient said 'not interested' but classified as wrong_number

Agent gave medical advice -- AI agent violated guardrail by offering medical guidance

Data capture errors -- Responses recorded incorrectly despite patient giving correct answers

Dataset for Problem 1

Train: 689 calls (59 tickets) -- use for model development

Validation: 144 calls (11 tickets) -- use for hyperparameter tuning

Test: 159 calls -- **held out, unseen** -- used for final scoring (labels hidden)

Evaluation Metrics

Primary: **F1 Score** (harmonic mean of precision and recall)

Secondary: **Recall** (catching all bad calls matters more than false alarms)

Tertiary: **Precision** (reducing false positives saves reviewer time)

Bonus: Predict **ticket category** (which module caused the issue) for extra points.

Suggested Approaches

- Rule-based heuristics (duration vs response count, outcome patterns)
- Classical ML (XGBoost, Random Forest on engineered features)
- NLP on transcripts (embeddings + classifier, fine-tuned language models)
- Ensemble of heuristic + ML + LLM evaluation
- Anomaly detection (isolation forest, autoencoders on call embeddings)



Problem 2: AI Voice Agent Simulator

Build a voice agent that can handle patient medication refill check-in calls.

Objective

Using the call transcripts as reference, build an AI voice agent (using any realtime voice model -- OpenAI Realtime, ElevenLabs, Gemini Live, etc.) that can conduct a medication refill check-in call. The agent should ask 14 health questionnaire questions, handle natural conversation patterns, and correctly capture patient responses.

What the Agent Must Do

- Greet patient by name and confirm identity
- Ask 14 health check-in questions (weight, side effects, allergies, medications, etc.)
- Handle patients who go off-script (pricing questions, dosage concerns, side effect reports)
- Correctly capture and structure all responses
- Know when to escalate vs continue
- Handle edge cases: patient wants to reschedule, patient opts out, wrong number

Dataset for Problem 2

transcript_samples.json -- 35 representative call transcripts across all outcomes. Study these to understand the conversation flow, question patterns, and edge cases. You may also reference the full training dataset for additional examples.

Note: The training data's Q&A responses show the expected output format. Your agent should produce structured responses in a similar format.

Evaluation Criteria

Criteria	Description
Conversation Quality (30%)	Natural flow, appropriate responses, handles interruptions
Response Accuracy (30%)	Correctly captures all 14 questionnaire answers
Edge Case Handling (20%)	Pricing questions, reschedules, opt-outs, escalations
Technical Implementation (10%)	Code quality, architecture, documentation
Demo & Presentation (10%)	Live demo of agent handling a simulated call



Dataset Description

Files Provided

Folder	File	Contents	Purpose
csv/ json/ parquet/	hackathon_train.*	689 calls, 59 tickets	Model training
csv/ json/ parquet/	hackathon_val.*	144 calls, 11 tickets	Tuning
csv/ json/ parquet/	hackathon_test.*	159 calls, labels hidden	Final scoring
root	transcript_samples.json	35 transcripts	Voice agent ref
root	DATA_DICTIONARY.md	Field descriptions	Schema reference

Held-out test set (159 calls) is kept by organizers for final scoring. Participants receive test call features but labels are hidden.

Data Split Balance

Split	Calls	Tickets	Ticket Rate
Train	689	59	8.6%
Val	144	11	7.6%
Test (held out)	159	Hidden	--

Outcome Distribution (Train + Val)

Outcome	Train (tickets)	Val (tickets)	Total
completed	198 (23)	42 (5)	240
incomplete	155 (16)	32 (3)	187
opted_out	132 (7)	28 (1)	160
scheduled	89 (0)	19 (0)	108
escalated	64 (8)	13 (1)	77
wrong_number	40 (5)	8 (1)	48
voicemail	11 (0)	2 (0)	13



Key Fields Reference

Call Features

Field	Description
outcome	Call result: completed, incomplete, opted_out, scheduled, escalated, wrong_number, voicemail
call_duration	Duration in seconds
transcript_text	Full conversation transcript ([AGENT]: ... [USER]: ...)
responses_json	Structured Q&A -- 14 {question, answer} pairs from health check-in
whisper_mismatch_count	STT disagreements between two transcription systems
response_completeness	Fraction of questions answered (0.0 - 1.0)
validation_notes	Post-call AI validation analysis
whisper_transcript	Raw STT transcript (no role markers)

Target Variables

Field	Type	Description
has_ticket	bool	PRIMARY TARGET -- was a review ticket raised?
ticket_initial_notes	string	What the reviewer flagged
ticket_cat_*	string	Module-level diagnosis: " or 'issue'

Rules & Submission

- Teams of 1-4 participants
- You may use any programming language, framework, or pre-trained model
- For Problem 1: submit a CSV with columns [call_id, predicted_ticket (True/False)] for the test set
- For Problem 2: submit code + a 5-minute live demo of your voice agent
- No manual labeling of test data -- automated approaches only
- All code must be original work created during the hackathon

Prizes

Problem 1 winner: Highest F1 on held-out test set

Problem 2 winner: Best overall evaluation score (panel judged)

Best Innovation: Most creative or novel approach across either problem